Report – ML based portfolio management

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Introduction

In this project, we study the problem of allocating capital across a set of financial assets under a trade-off between expected return and allocation risk. We adopt the Markowitz mean-variance framework as the main theoretical basis, where the risk is measured by the variance of portfolio returns, and we investigate how portfolio construction changes when expected returns and covariance matrices are estimated from historical data rather than assumed known.
In part 1, we present the theoretical foundations of the Markowitz problem and derive the efficient frontier, both in the unconstrained case and under realistic long-only constraints. Part 2 discusses statistical estimation of returns and covariance matrices, including shrinkage and time-weighted methods, as well as the limits of the Gaussian assumption. Part 3 details the implementation in Python, the backtesting protocol and the empirical results on real market data, complemented by an interactive dashboard to make the evaluation of results more concrete.

1 Part 1: Theoretical Foundations

Modern Portfolio Theory (MPT), introduced by Harry Markowitz in his seminal 1952 paper *Portfolio Selection*, provides a mathematical framework for constructing portfolios that maximize expected return for a given level of risk. The central insight is that diversification across assets reduces portfolio risk, measured by variance, without necessarily sacrificing expected return.

We consider a universe of n assets. The return of asset i is modeled as a random variable X_i, for $1 \leq i \leq n$. Under the Markowitz framework, returns are assumed to be jointly Gaussian, so each asset is fully characterized by its expected return and its covariance with other assets.

The empirical mean return of asset i, estimated over a sample of T observations from t_0 to t_f, is:

$$\mu_i = \frac{1}{T} \sum_{t=t_0}^{t_f} r_{i,t}$$

The empirical covariance matrix $C \in \mathcal{M}_n(\mathbb{R})$, which captures both individual asset variances and pairwise co-movements, is defined by:

$$C_{ij} = \frac{1}{T-1} \sum_{t=t_0}^{t_f} (r_{i,t} - \mu_i)(r_{j,t} - \mu_j)$$

C is symmetric positive semi-definite, its diagonal entries $C_{ii} = \sigma_i^2$ represent the variance of each individual asset, while off-diagonal entries capture the linear dependence between assets, normalized as $C_{ij} = \rho_{ij}\sigma_i\sigma_j$, where ρ_{ij} is the correlation coefficient between assets i and j. A portfolio is defined by a weight vector $\mathbf{w} = (w_1, \ldots, w_n)^T \in \mathbb{R}^n$, where w_i denotes the fraction of wealth allocated to asset i. The weights satisfy the budget constraint:

$$\mathbf{w}^T \mathbf{1} = 1, \qquad \text{where } \mathbf{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$$

Under this allocation, the portfolio return is the weighted average of individual asset returns, with expected value $\mathbf{w}^T \boldsymbol{\mu}$ and variance $\mathbf{w}^T C\, \mathbf{w}$. The investor seeks to minimize portfolio variance for a targeted expected return $\mu^* \in \mathbb{R}$. This yields the following quadratic program:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^T C\, \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T \mathbf{1} = 1, \\ & \mathbf{w}^T \boldsymbol{\mu} = \mu^*. \end{aligned}$$

This is a convex quadratic optimization problem with linear constraints, which admits a closed-form solution via Lagrange multipliers. By varying the target return μ^*, one traces out the **efficient frontier**: the set of portfolios that achieve the minimum variance for each level of expected return. Rational investors will only hold portfolios lying on this frontier.

To solve the optimization problem, we introduce the Lagrangian with two multipliers $\lambda, \gamma \in \mathbb{R}$:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \mathbf{w}^T C\, \mathbf{w} - \lambda \left(\mathbf{w}^T \boldsymbol{\mu} - \mu^*\right) - \gamma \left(\mathbf{w}^T \mathbf{1} - 1\right)$$

Taking the first-order condition with respect to $\mathbf{w}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 2C\mathbf{w} - \lambda \boldsymbol{\mu} - \gamma \mathbf{1} = 0$$

Assuming C is invertible (which holds when no asset is a linear combination of others), we obtain:

$$\mathbf{w}^* = \frac{1}{2} C^{-1} (\lambda \boldsymbol{\mu} + \gamma \mathbf{1})$$

The multipliers λ and γ are determined by substituting $\mathbf{w}^*$ back into the two constraints.

Constraint 1: $\mathbf{w}^{*T} \boldsymbol{\mu} = \mu^*$

$$\mathbf{w}^{*T} \boldsymbol{\mu} = \frac{1}{2} (\lambda \boldsymbol{\mu} + \gamma \mathbf{1})^T C^{-1} \boldsymbol{\mu} = \frac{1}{2} \left(\lambda \boldsymbol{\mu}^T C^{-1} \boldsymbol{\mu} + \gamma\, \mathbf{1}^T C^{-1} \boldsymbol{\mu}\right) = \frac{1}{2} (\lambda A + \gamma B) = \mu^*$$

Constraint 2: $\mathbf{w}^{*T}\mathbf{1} = 1$

$$\mathbf{w}^{*T}\mathbf{1} = \frac{1}{2}\left(\lambda\boldsymbol{\mu} + \gamma\mathbf{1}\right)^T C^{-1}\mathbf{1} = \frac{1}{2}\left(\lambda\boldsymbol{\mu}^T C^{-1}\mathbf{1} + \gamma\,\mathbf{1}^T C^{-1}\mathbf{1}\right) = \frac{1}{2}\left(\lambda B + \gamma D\right) = 1$$

We defined the following scalar quantities:

$$A = \boldsymbol{\mu}^T C^{-1}\boldsymbol{\mu}, \qquad B = \boldsymbol{\mu}^T C^{-1}\mathbf{1} = \mathbf{1}^T C^{-1}\boldsymbol{\mu}, \qquad D = \mathbf{1}^T C^{-1}\mathbf{1}$$

and the determinant $\Delta = AD - B^2 > 0$ (positive by the Cauchy-Schwarz inequality, assuming $\boldsymbol{\mu}$ and $\mathbf{1}$ are not collinear).

Injecting $\mathbf{w}^*$ into $\mathbf{w}^{*T}\boldsymbol{\mu} = \mu^*$ and $\mathbf{w}^{*T}\mathbf{1} = 1$ yields the linear system:

$$\frac{1}{2}\begin{pmatrix} A & B \\ B & D \end{pmatrix}\begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} \mu^* \\ 1 \end{pmatrix}$$

Inverting this 2×2 system:

$$\lambda = \frac{2(D\mu^* - B)}{\Delta}, \qquad \gamma = \frac{2(A - B\mu^*)}{\Delta}$$

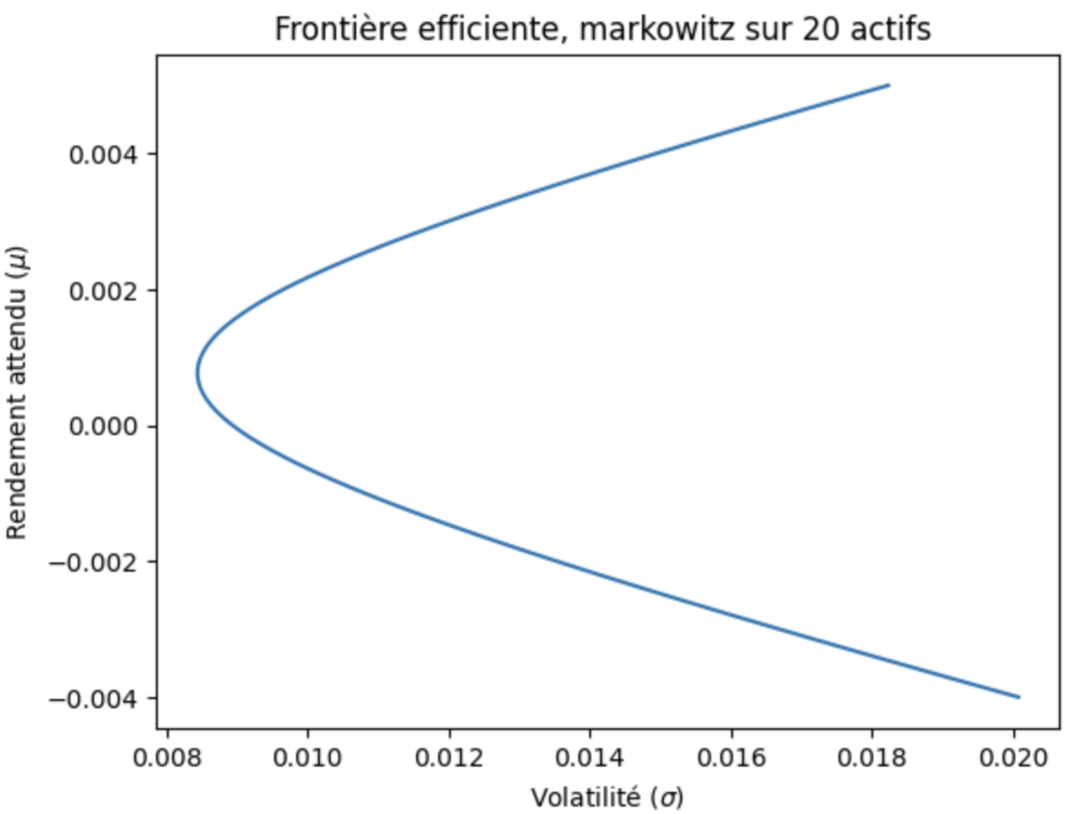

Figure 1: Drawing the efficient boundary for the 20 assets

Substituting λ and γ back into the expression for $\mathbf{w}^*$:

$$\boxed{\mathbf{w}^* = \frac{D\mu^* - B}{\Delta} C^{-1}\boldsymbol{\mu} + \frac{A - B\mu^*}{\Delta} C^{-1}\mathbf{1}}$$

This can be rewritten as a convex combination of two reference portfolios:

$$\mathbf{w}^* = \alpha\, \mathbf{w}_{\boldsymbol{\mu}} + (1-\alpha)\, \mathbf{w}_{\mathbf{1}}$$

where:

$$\mathbf{w}_{\boldsymbol{\mu}} = \frac{C^{-1}\boldsymbol{\mu}}{\mathbf{1}^T C^{-1}\boldsymbol{\mu}}, \qquad \mathbf{w}_{\mathbf{1}} = \frac{C^{-1}\mathbf{1}}{\mathbf{1}^T C^{-1}\mathbf{1}}, \qquad \alpha = \frac{D\mu^* - B}{\Delta} \cdot B$$

This is the **two-fund separation theorem**: any efficient portfolio is a linear combination of just two fundamental portfolios, regardless of the target return μ^*. By symmetry of the optimization problem, one can instead fix the portfolio variance $\sigma^{*2} = \mathbf{w}^T C\mathbf{w}$ and seek the allocation that maximizes the expected return $\mathbf{w}^T\boldsymbol{\mu}$:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mathbf{w}^T\boldsymbol{\mu} \\ \text{subject to} \quad & \mathbf{w}^T C\mathbf{w} = \sigma^{*2}, \\ & \mathbf{w}^T\mathbf{1} = 1. \end{aligned}$$

The Lagrangian, with multiplier η on the (now nonlinear) variance constraint and γ on the budget constraint, is:

$$\mathcal{L}(\mathbf{w}, \eta, \gamma) = \mathbf{w}^T\boldsymbol{\mu} - \eta\left(\mathbf{w}^T C\mathbf{w} - \sigma^{*2}\right) - \gamma\left(\mathbf{w}^T\mathbf{1} - 1\right)$$

Setting the gradient with respect to $\mathbf{w}$ to zero:

$$\boldsymbol{\mu} - 2\eta C\mathbf{w} - \gamma\mathbf{1} = 0 \implies \mathbf{w}^* = \frac{1}{2\eta} C^{-1}\left(\boldsymbol{\mu} - \gamma\mathbf{1}\right)$$

This expression has exactly the same structure as in the variance-minimization problem (it is again a linear combination of $C^{-1}\boldsymbol{\mu}$ and $C^{-1}\mathbf{1}$), which confirms that both formulations trace out the **same efficient frontier**: minimizing variance for a given return and maximizing return for a given variance are two dual descriptions of the same set of optimal portfolios. In practice, η and γ are pinned down by substituting $\mathbf{w}^*$ back into the budget and variance constraints, exactly as in the previous derivation, leading to the same expression of $\mu^*(\sigma^{*2})$ obtained by inverting the hyperbola $\sigma^2(\mu^*) = \dfrac{D\mu^{*2} - 2B\mu^* + A}{\Delta}$.

However, investors are sometimes prevented, by mandate or regulation, from taking short positions, that is, selling a stock they do not currently own. This corresponds to forbidding negative coordinates in $\mathbf{w}$, and the long-only constraint is imposed as:

$$w_i \geq 0, \quad \forall i \in \{1, \ldots, n\}.$$

Complete Solution with the Long-Only Constraint via KKT

We solve explicitly:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^T C\, \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T\mathbf{1} = 1, \\ & \mathbf{w}^T\boldsymbol{\mu} = \mu^*, \\ & w_i \geq 0, \quad \forall i \in \{1, \ldots, n\}. \end{aligned}$$

Lagrangian. Introduce multipliers $\lambda, \gamma \in \mathbb{R}$ for the equality constraints and $\nu_i \geq 0$ for each inequality constraint $-w_i \leq 0$:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma, \boldsymbol{\nu}) = \mathbf{w}^T C\mathbf{w} - \lambda\left(\mathbf{w}^T\boldsymbol{\mu} - \mu^*\right) - \gamma\left(\mathbf{w}^T\mathbf{1} - 1\right) - \boldsymbol{\nu}^T\mathbf{w}$$

KKT conditions. Since the objective is convex and the constraints are affine, the KKT conditions are necessary and sufficient for global optimality:

$$\begin{aligned}
&\text{(stationarity)} \qquad 2C\mathbf{w}^* - \lambda\boldsymbol{\mu} - \gamma\mathbf{1} - \boldsymbol{\nu} = 0 \\
&\text{(primal feasibility)} \qquad {\mathbf{w}^*}^T\mathbf{1} = 1, \quad {\mathbf{w}^*}^T\boldsymbol{\mu} = \mu^*, \quad w_i^* \geq 0 \;\; \forall i \\
&\text{(dual feasibility)} \qquad \nu_i \geq 0 \;\; \forall i \\
&\text{(complementary slackness)} \qquad \nu_i \, w_i^* = 0 \;\; \forall i
\end{aligned}$$

Partition of assets. Let $\mathcal{I} = \{i : w_i^* > 0\}$ be the set of active (held) assets and $\mathcal{J} = \{i : w_i^* = 0\}$ the set of excluded assets, with $\mathcal{I} \cup \mathcal{J} = \{1, \ldots, n\}$.

By complementary slackness, $\nu_i = 0$ for all $i \in \mathcal{I}$. Restricting the stationarity condition to the active set, and writing $C_{\mathcal{I}}$, $\boldsymbol{\mu}_{\mathcal{I}}$, $\mathbf{1}_{\mathcal{I}}$ for the submatrix and subvectors restricted to $\mathcal{I}$:

$$2C_{\mathcal{I}}\mathbf{w}_{\mathcal{I}}^* = \lambda\boldsymbol{\mu}_{\mathcal{I}} + \gamma\mathbf{1}_{\mathcal{I}}$$

This is exactly the unconstrained stationarity equation, but restricted to the sub-universe $\mathcal{I}$. Assuming $C_{\mathcal{I}}$ is invertible:

$$\boxed{\mathbf{w}_{\mathcal{I}}^* = \frac{1}{2}C_{\mathcal{I}}^{-1}\left(\lambda\boldsymbol{\mu}_{\mathcal{I}} + \gamma\mathbf{1}_{\mathcal{I}}\right), \qquad w_j^* = 0 \;\; \forall j \in \mathcal{J}}$$

with λ, γ pinned down, exactly as before, by substituting into the two equality constraints restricted to $\mathcal{I}$:

$$A_{\mathcal{I}} = \boldsymbol{\mu}_{\mathcal{I}}^T C_{\mathcal{I}}^{-1}\boldsymbol{\mu}_{\mathcal{I}}, \quad B_{\mathcal{I}} = \boldsymbol{\mu}_{\mathcal{I}}^T C_{\mathcal{I}}^{-1}\mathbf{1}_{\mathcal{I}}, \quad D_{\mathcal{I}} = \mathbf{1}_{\mathcal{I}}^T C_{\mathcal{I}}^{-1}\mathbf{1}_{\mathcal{I}}$$

$$\lambda = \frac{2(D_{\mathcal{I}}\mu^* - B_{\mathcal{I}})}{\Delta_{\mathcal{I}}}, \qquad \gamma = \frac{2(A_{\mathcal{I}} - B_{\mathcal{I}}\mu^*)}{\Delta_{\mathcal{I}}}, \qquad \Delta_{\mathcal{I}} = A_{\mathcal{I}}D_{\mathcal{I}} - B_{\mathcal{I}}^2$$

Validity check. This candidate solution is optimal if and only if two conditions hold simultaneously:

$$w_i^* > 0 \;\; \forall i \in \mathcal{I}, \qquad \nu_j = -\left(2(C\mathbf{w}^*)_j - \lambda\mu_j - \gamma\right) \geq 0 \;\; \forall j \in \mathcal{J}$$

The first condition checks that the held assets indeed have positive weight; the second checks that excluding the remaining assets is consistent with the optimality conditions (their "implicit return" is dominated).

Resolution method. The KKT analysis above theoretically characterizes the long-only optimum: the solution is the unconstrained two-fund formula restricted to an a priori unknown active set $\mathcal{I}$. This set can in principle be found via the classical *active-set algorithm* for quadratic programming: start with $\mathcal{I} = \{1, \ldots, n\}$, compute $\mathbf{w}^*$ from the unconstrained formula, drop the most negative weight(s), and repeat on the reduced set until all weights are non-negative.

We do not run this loop explicitly, however. The closed-form Lagrangian solution is used only for the equality-constrained case (budget and target return); the long-only problem, and more generally any problem with caps $w_i \leq w_{\max}$ or transaction-cost penalties, is handed directly to a general convex quadratic-programming solver (`CVXPY`, `CLARABEL` backend, see Part 3). Since the objective is convex and all constraints are affine, this solver returns the exact global optimum characterized by the KKT conditions above, without our enumerating the active set by hand. The active-set derivation thus explains *why* the long-only frontier is piecewise-quadratic (it is the two-fund formula applied recursively on shrinking subsets of assets), while the convex solver is what computes it in practice.

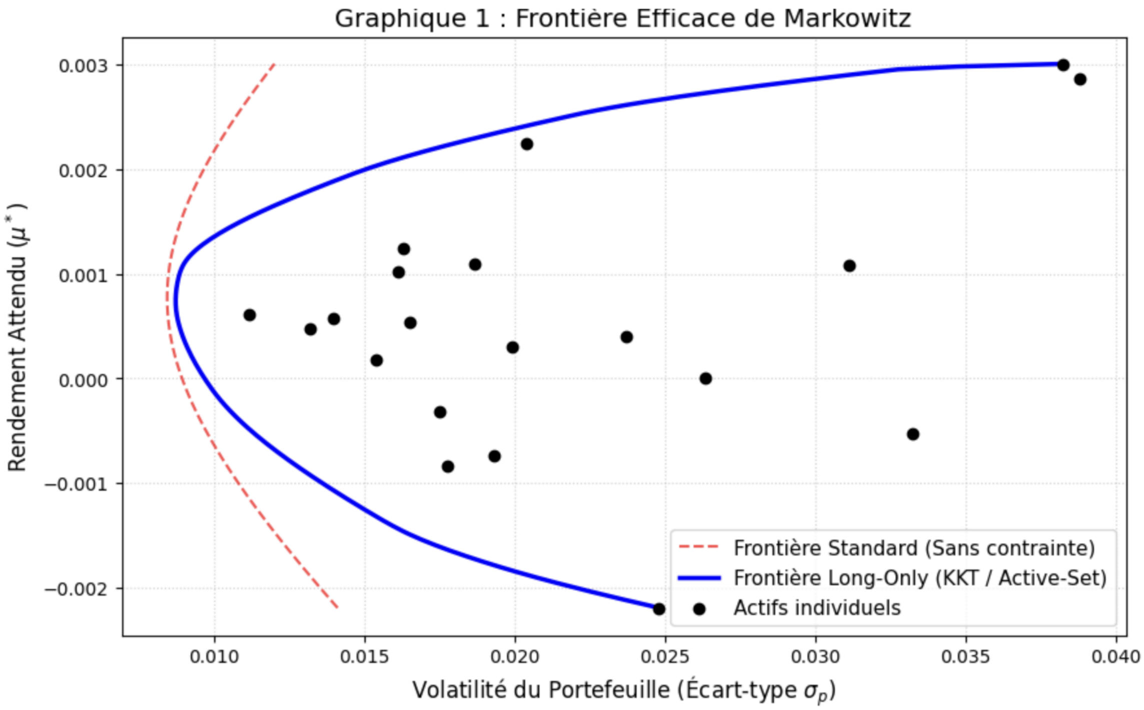

Figure 2: Efficient frontier with long-only compared to the previous version

2 Part 2: Probability and statistics

Under Markowitz framework, the solution is optimal if and only if asset returns are Gaussian and the mean vector $\boldsymbol{\mu}$ and the covariance matrix C are perfectly known beforehand. In practice, however, these parameters are never directly observable: an investor only has access to a finite history of past returns, from which $\boldsymbol{\mu}$ and C must be estimated statistically. This estimation step is a critical, and often underappreciated, source of error in the mean-variance framework, since the entire optimization is built on top of these estimated parameters rather than the true ones.
We keep the same universe of n assets and Gaussian i.i.d. return model introduced in Part 1, with empirical mean return μ_i and empirical covariance matrix $C \in \mathcal{M}_n(\mathbb{R})$. Here we focus on how the mean vector $\boldsymbol{\mu}$ and the covariance matrix C of these returns are statistically estimated from historical data.

While straightforward to implement, this naive sample-based approach suffers from several limitations, which directly affect the reliability of the resulting Markowitz allocation:

- **Estimation error on $\boldsymbol{\mu}$.** The sample mean is known to be a particularly noisy estimator of expected returns: its standard error scales as $\sigma_i/\sqrt{T}$, which decreases only slowly with the sample size. Since portfolio weights in the Markowitz solution depend linearly on $C^{-1}\boldsymbol{\mu}$, small estimation errors in $\boldsymbol{\mu}$ can translate into large and unstable swings in the optimal allocation (*error maximization* phenomenon).
- **Ill-conditioning of C.** When the number of assets n is large relative to the number of observations T (a common situation when n approaches or exceeds T), the sample covariance matrix becomes ill-conditioned or even singular, making its inversion numerically unstable and the resulting weights extremely sensitive to small perturbations in the data.
- **Non-stationarity.** The i.i.d. assumption implies that $\boldsymbol{\mu}$ and C are constant over the estimation window, which is rarely true in practice: volatility tends to cluster over time, and correlations between assets are known to shift, particularly during periods of market stress, when diversification benefits tend to disappear precisely when they are most needed.

These limitations motivate the use of more robust estimation techniques.

To address the non-stationarity limitation of the naive estimator, the **Exponentially Weighted Moving Average (EWMA)** method assigns exponentially decreasing weights to past observations, rather than weighting all T historical returns equally. Recent observations, more informative about the current market regime, thus contribute more to the estimate than distant ones. The weight assigned to the observation at time t, within a window ending at T, is:

$$w_t = (1-\lambda)\,\lambda^{T-1-t}$$

where $\lambda \in (0,1)$ is the decay factor controlling how quickly the influence of past observations fades. The corresponding covariance estimator is:

$$\hat{\Sigma} = X_w^T X_w$$

where X_w is the matrix of returns weighted by $\sqrt{w_t}$, i.e. each return row $\mathbf{r}_t$ is scaled by $\sqrt{w_t}$ before computing the cross products. Following the standard RiskMetrics convention, we do *not* subtract a sample mean before forming these products: at the daily horizon the mean return is negligible relative to volatility ($|\mu_i| \ll \sigma_i$), so the estimator effectively targets the (non-centered) second moment $\mathbb{E}[\mathbf{r}\mathbf{r}^T]$, which coincides with the covariance up to a $\mathcal{O}(\mu\mu^T)$ term. Writing $\hat{\Sigma}$ as a Gram matrix $X_w^T X_w$ also guarantees that it remains PSD by construction, an important property since the Markowitz solution requires inverting (or factorizing) this matrix. The decay factor λ directly controls the trade-off between responsiveness and stability: a value close to 1 behaves similarly to the naive equal-weighted estimator, while a lower value makes the estimate highly reactive to recent shocks, at the cost of higher estimation noise. The RiskMetrics methodology, widely used in practice, recommends:

$$\lambda = 0.94 \text{ (daily data)}, \qquad \lambda = 0.97 \text{ (monthly data)}$$

The practical interpretation of λ is best captured by the *effective half-life* of the weighting scheme, i.e. the number of periods after which an observation's weight has been reduced by half:

$$\text{half-life} = \frac{\ln 2}{\ln(1/\lambda)} \approx 11 \text{ days } (\lambda = 0.94)$$

This makes EWMA particularly well suited to capturing volatility clustering and shifting correlations, directly addressing the non-stationarity concern raised above, at the cost of introducing an additional hyperparameter λ that must itself be calibrated or chosen by convention.

To fix ill-conditioning, the **Ledoit-Wolf shrinkage** estimator regularizes the sample covariance matrix by pulling it towards a more stable target matrix F (for instance, a diagonal matrix, or a matrix implied by a single-factor or constant-correlation model). It takes the form:

$$\hat{\Sigma}_{\text{shrink}} = \delta F + (1-\delta)\, C$$

where $\delta \in [0, 1]$ is the shrinkage intensity, chosen to minimize the expected quadratic loss to the true covariance. This deliberately introduces a small bias toward the target F, in exchange for a substantial reduction in the variance of the estimator. The resulting matrix is better conditioned than the raw sample covariance, its smallest eigenvalues are pulled upward and its largest eigenvalues are pulled downward, which makes its inversion numerically stable even when n approaches or exceeds T, a regime in which the naive sample covariance matrix is singular or near-singular. In the context of Markowitz optimization, this directly mitigates the error maximization problem: since the optimal weights depend on C^{-1}, a better conditioned and more stable covariance estimate translates into less extreme and more robust portfolio allocations.

EWMA and Ledoit-Wolf shrinkage address two distinct, complementary weaknesses of the naive empirical estimator: the former tackles the assumption of time-invariant parameters by adapting to recent market conditions, while the latter tackles the curse of dimensionality and numerical instability by regularizing the covariance structure itself. In practice, the two approaches are often combined, for instance by applying Ledoit-Wolf shrinkage to an EWMA-weighted covariance matrix, in order to obtain an estimator that is both responsive to changing market regimes and numerically well behaved.

In the following figure, we plot the correlation matrices obtained from the empirical (sample) covariance, a rolling window of 30 days, the EWMA method, and the Ledoit-Wolf method. κ denotes the condition number of the covariance matrix. A value of κ close to 1 indicates that the matrix is well-conditioned, whereas a high value indicates that it is poorly conditioned.
We use the daily returns of 8 stocks between 2024 and 2026.

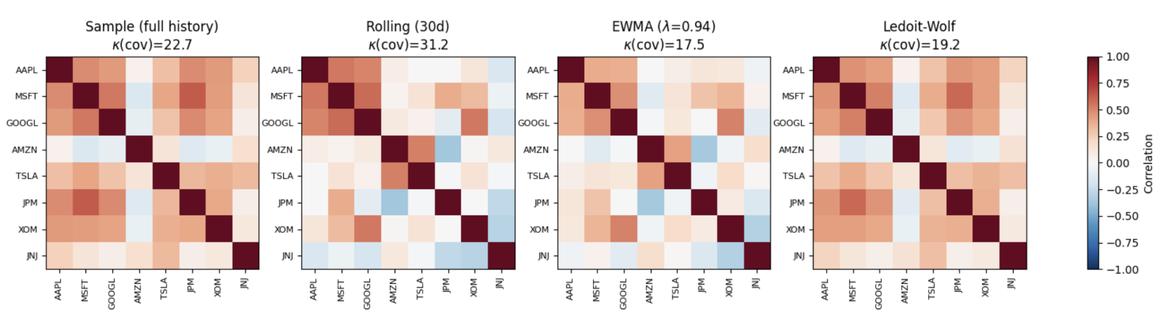

Figure 3: The comparison of the 4 correlation matrices

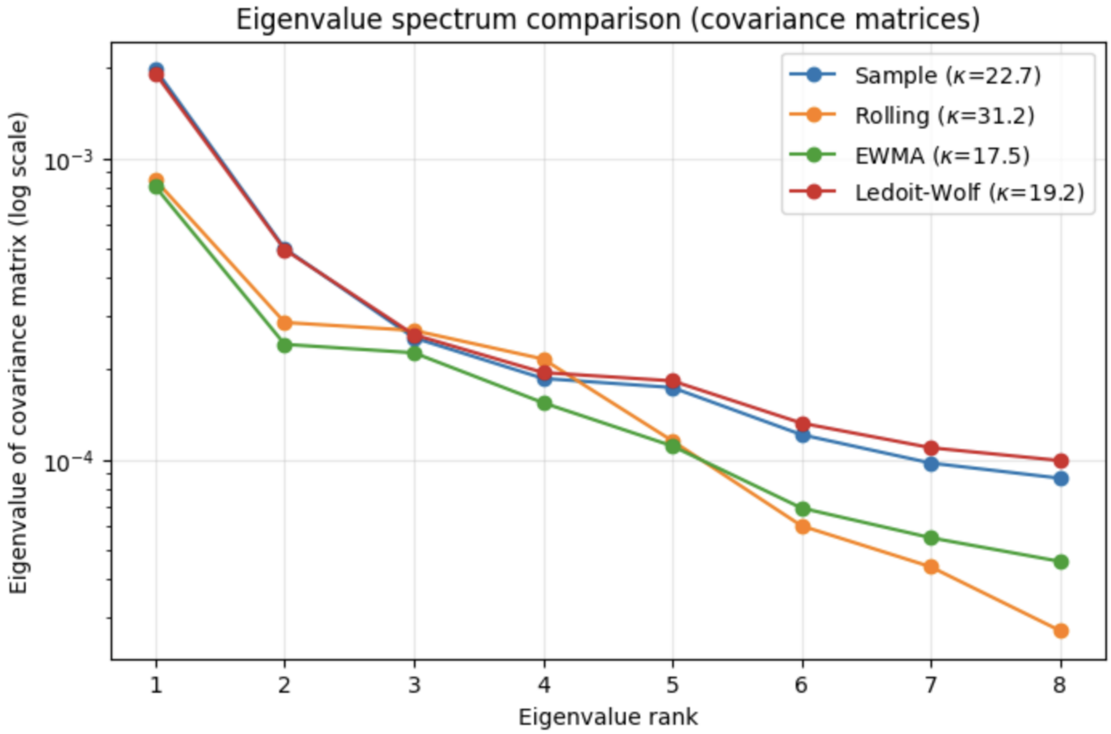

Figure 4: The comparison of the eigenvalues

As we did for the covariance matrix we can also improve the estimation of returns. The sample mean $\hat{\boldsymbol{\mu}} = \frac{1}{T}\sum_t \mathbf{r}_t$ is an unbiased estimator of the true mean vector $\boldsymbol{\mu}$, but it is very noisy in practice, particularly when the number of assets n is large relative to the number of observations T. A natural solution is to shrink the sample mean toward a common target, exactly as Ledoit-Wolf shrinks the covariance matrix. The **James-Stein estimator** does precisely this: instead of trusting each asset's sample mean individually, it pulls all estimates toward a common reference value μ_0, typically chosen as the grand mean of all assets:

$$\mu_0 = \frac{1}{n}\sum_{i=1}^{n}\hat{\mu}_i$$

The estimator then takes the form:

$$\hat{\mu}_i^{\mathrm{JS}} = (1-\delta)\,\hat{\mu}_i + \delta\,\mu_0$$

where $\delta \in [0,1]$ controls how strongly each individual estimate is pulled toward the common mean μ_0: $\delta = 0$ recovers the raw sample mean, $\delta = 1$ assigns the same expected return μ_0 to all assets.
Why does this work? The intuition is straightforward: in a universe of n assets, the sample mean of each asset is estimated from only T observations and therefore contains a large amount of noise. Some assets will have an artificially high sample mean simply by chance, and others an artificially low one. By shrinking all estimates toward a common value, the James-Stein estimator reduces these extreme, noise-driven deviations, while introducing a small bias (*Stein's phenomenon*) for any true $\boldsymbol{\mu}$ when $n \geq 3$.

$$\mathbb{E}\left[\|\hat{\boldsymbol{\mu}}^{\mathrm{JS}} - \boldsymbol{\mu}\|^2\right] < \mathbb{E}\left[\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2\right]$$

In the portfolio context, this directly translates into more stable and less extreme optimal weights $\mathbf{w}^* \propto C^{-1}\boldsymbol{\mu}$, since the input $\boldsymbol{\mu}$ fed into the optimizer is less contaminated by estimation noise. The James-Stein estimator is therefore the natural complement to Ledoit-Wolf shrinkage: the former regularizes the mean vector $\boldsymbol{\mu}$, the latter regularizes the covariance matrix C, and the two are typically combined in practice to obtain a fully robust Markowitz optimizer.

Limitations of the Gaussian Assumption: Higher Moments and Risk Indicators

The entire framework developed above relies on the assumption that asset returns are Gaussian, fully characterized by their first two moments, the mean μ_i and the variance σ_i^2. In practice, however, financial returns frequently exhibit departures from normality, in particular fat tails (extreme events occurring more often than a Gaussian model would predict) and asymmetry between gains and losses. Several indicators are used to detect and quantify these deviations.

Skewness measures the asymmetry of the return distribution around its mean, with negative values indicating heavier left tails. It is estimated empirically as the standardized third central moment:

$$\text{Skew}_i = \frac{\frac{1}{T}\sum_{t=t_0}^{t_f}(r_{i,t}-\mu_i)^3}{\sigma_i^3}$$

Large losses are rare but more extreme than large gains, a pattern commonly observed in equity returns and particularly undesirable for risk-averse investors.

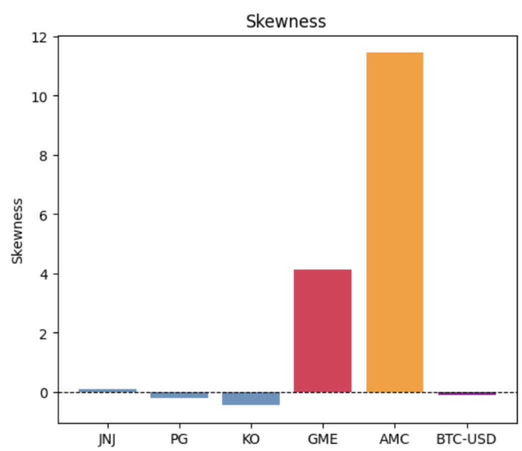

Figure 5: The skewness for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are highly non-gaussian

Kurtosis captures tail thickness relative to a Gaussian distribution. It is estimated as the standardized fourth central moment:

$$\text{Kurt}_i = \frac{\frac{1}{T}\sum_{t=t_0}^{t_f}(r_{i,t}-\mu_i)^4}{\sigma_i^4}$$

A Gaussian distribution has a kurtosis of exactly 3. Financial returns typically display *excess kurtosis* or leptokurtosis (Kurt > 3), meaning extreme returns occur more frequently than under normality.

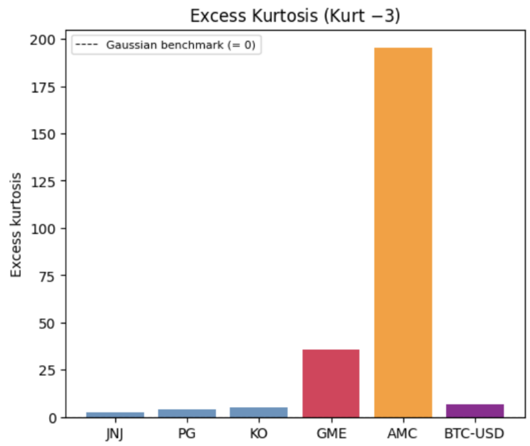

Figure 6: The leptokurtosis coefficient for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are highly non-gaussian

The Gini coefficient, more commonly used to measure inequality in income distributions, provides an alternative measure of dispersion based on pairwise absolute differences, less tied to the Gaussian assumption than variance:

$$\text{Gini}_i = \frac{1}{2T^2\mu_i} \sum_{s=t_0}^{t_f} \sum_{t=t_0}^{t_f} |r_{i,s} - r_{i,t}|$$

A higher Gini coefficient indicates a more dispersed and unequal distribution of returns, capturing extreme inequality between observations in a way variance alone does not.

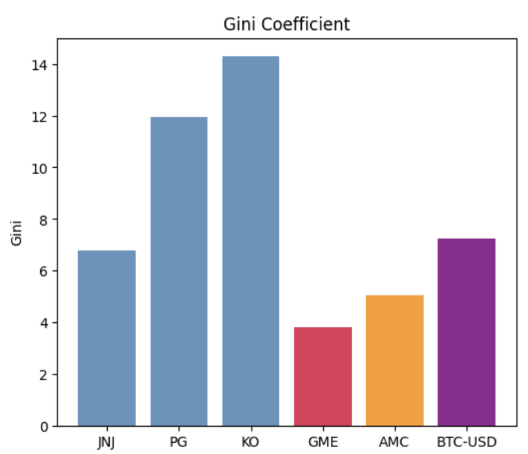

Figure 7: The Gini coefficient for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are absolutely non-gaussian

Taken together, these indicators allow the analyst to flag assets whose return distribution departs significantly from normality, which directly threatens the validity of the mean-variance framework: when skewness and excess kurtosis are large, variance no longer fully captures the risk faced by the investor, since it ignores the asymmetry and the probability of extreme losses.

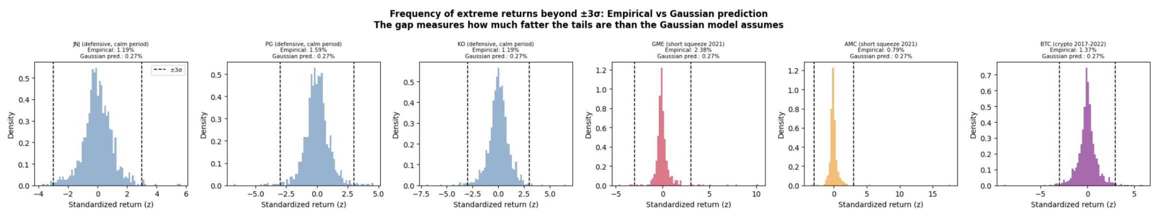

Figure 8: The distribution of return for JNJ,PG,KO,GME,AMC,BTC where GME,AMC,BTC are absolutely non-gaussian

Value at Risk as a Complementary Risk Measure

To address this limitation, the **Value at Risk (VaR)** is commonly introduced as a risk measure that directly targets the tail of the distribution rather than its overall dispersion. For a confidence level α (typically 95% or 99%), the VaR of a portfolio over a holding period is defined as the loss threshold that will not be exceeded with probability α:

$$\mathbb{P}\left(R_p \leq -\mathrm{VaR}_\alpha\right) = 1 - \alpha$$

where $R_p = \mathbf{w}^T\mathbf{r}$ is the portfolio return. Under the Gaussian assumption, VaR_α has a simple closed form, $\mathrm{VaR}_\alpha = -\mathbf{w}^T\boldsymbol{\mu}+z_\alpha\sqrt{\mathbf{w}^T C\mathbf{w}}$ where z_α is the corresponding quantile of the standard normal distribution. However, when returns exhibit fat tails or skewness, this Gaussian VaR systematically underestimates the true risk of large losses, and the VaR is instead estimated empirically (historical simulation) or via Monte Carlo simulation from the actual, non-Gaussian distribution of returns.

In portfolio allocation, VaR is used in two main ways. First, as a constraint added to the classical mean-variance problem, the investor minimizes variance (or maximizes return) subject to the additional condition that the portfolio's VaR does not exceed a maximum acceptable threshold VaR^*, which prevents the optimizer from selecting portfolios that look efficient in mean-variance terms but carry a disproportionate risk of extreme loss. Second, as an alternative objective function, replacing the variance entirely: the investor minimizes $\mathrm{VaR}_\alpha(\mathbf{w})$ directly, subject to the budget and return constraints, which better accounts for tail risk but generally loses the closed-form tractability of the Markowitz solution, since VaR is not, in general, a convex function of $\mathbf{w}$ (a limitation that motivates the use of the related but convex Conditional Value at Risk, or Expected Shortfall, in more advanced allocation frameworks).

Monte Carlo Simulation for Efficient Frontier An alternative to solving the quadratic program analytically is to approximate the efficient frontier numerically via Monte Carlo: randomly generate many portfolios, compute their return and variance, and retain the dominant ones.

Procedure.

1. Generate N random weight vectors $\mathbf{w}^{(k)} \in \mathbb{R}^n$, $k = 1, \ldots, N$, typically via uniform or Dirichlet-distributed draws normalized so that $\mathbf{w}^{(k)T}\mathbf{1} = 1$ (Dirichlet is preferred since it also enforces $w_i \geq 0$, convenient for the long-only case).
2. Compute $\mu^{(k)} = \mathbf{w}^{(k)T}\boldsymbol{\mu}$ and $\sigma^{(k)2} = \mathbf{w}^{(k)T}C\,\mathbf{w}^{(k)}$ for each portfolio.
3. Plot all N portfolios in the (σ, μ) plane, filling the feasible region.
4. Identify the upper-left boundary, i.e. the highest-return portfolio per risk level (or lowest-risk per return level): this empirical boundary approximates the efficient frontier.

Accuracy. The method is intuitive and avoids matrix inversion, but it only *approximates* the true frontier, and from the inside: since sampled portfolios are random draws from the feasible set, the cloud's boundary always lies strictly below (in return) or to the right (in risk) of the true analytical frontier. The gap depends on:

- **Number of simulations N.** As $N \to \infty$ the cloud's boundary converges to the true frontier, but slowly: tracing it in an n-asset universe means sampling an $(n-1)$-dimensional simplex, and the chance of drawing a near-optimal portfolio drops sharply as n grows. A few thousand portfolios suffice for $n \leq 5$, while $n > 10$ needs tens or hundreds of thousands for a smooth, accurate boundary.

- **Sampling distribution.** Naive uniform sampling on the simplex concentrates near the center (equal weights), under-sampling the extreme, concentrated portfolios that often lie on the frontier itself, especially near its endpoints (minimum-variance and maximum-return portfolios), degrading precision unless the scheme is adapted.

For this reason, Monte Carlo simulation is mainly used as a **visualization and intuition-building tool** to illustrate the shape of the feasible set and the position of the efficient frontier as its boundary, rather than as a precise computational method. For an accurate or production-grade frontier, the closed-form solution (or its constrained quadratic programming counterpart) derived earlier remains strictly superior, since it locates the exact frontier without sampling error and at a negligible computational cost.

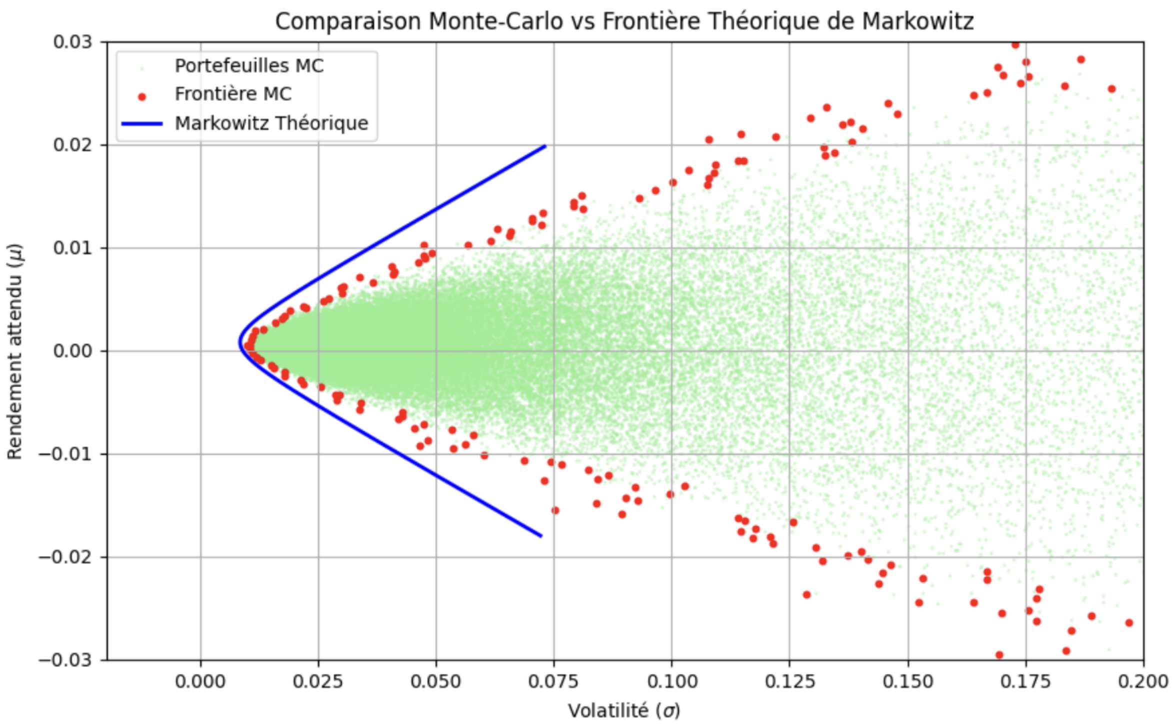

Figure 9: Plot of the theoretical efficient frontier and Monte Carlo simulated portfolios under Gaussian assumption

3 Part 3: Numerical Resolution and Representation

All numerical experiments presented in this report were implemented in Python. Historical price data was retrieved using the `yfinance` library, which provides access to Yahoo Finance market data. For each asset in the universe, we collected daily adjusted closing prices over the chosen estimation window, using the `yf.download()` function. Daily *simple* (arithmetic) returns were then computed from the closing price series using the `pct_change()` method, which computes for each day t:

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

We deliberately use simple returns rather than logarithmic returns, because the portfolio return aggregates *linearly across assets* as $r_{p,t} = \mathbf{w}^T \mathbf{r}_t = \sum_i w_i r_{i,t}$, which is exactly the quantity the mean-variance framework operates on; log returns, by contrast, aggregate linearly across *time* but not across assets, so $\mathbf{w}^T$ of log returns is not the log return of the portfolio. Missing values introduced by market closures or data gaps were removed using `dropna()`, ensuring that the return matrix fed into the optimizer is complete and well-defined.

Optimization method

From the cleaned return matrix, we estimated the mean vector $\boldsymbol{\mu}$ and the covariance matrix C using the different estimators described in the previous section: the naive sample estimator, the rolling window estimator, the EWMA estimator with $\lambda = 0.94$, and the Ledoit-Wolf shrinkage estimator. The James-Stein estimator was used for the mean vector. For each combination of estimators, the optimal Markowitz portfolio weights were computed by solving the quadratic program described in Section 1, either via the closed-form analytical solution derived using Lagrange multipliers, or via numerical quadratic programming solvers for the constrained long-only case.

A natural difficulty in evaluating portfolio performance is that asset prices tend to rise on average over long horizons, meaning that almost any invested portfolio will show positive cumulative returns simply due to the general upward trend of equity markets. A rising portfolio value is therefore not, by itself, evidence that the optimization model is adding value. To rigorously assess the effectiveness of each method, we therefore compare its performance against two reference benchmarks:

- **Equally-weighted portfolio**: each asset receives a weight $w_i = 1/n$, regardless of its return or risk. This naive allocation serves as a baseline representing the performance achievable with no optimization at all, and is known to be surprisingly competitive in practice.
- **S&P 500 index**: the performance of the broad U.S. equity market, representing the opportunity cost of simply holding a passive index fund rather than running an active mean-variance optimizer.

The comparison is conducted via **backtesting**.
To evaluate a portfolio method honestly, we must separate what is *known at decision time* from what happens *after* the decision. This is the core logic of our backtesting framework.

1) Walk-forward principle. At each rebalancing date t, the model is calibrated only on a past estimation window $[t - W, t)$, where W is the training length. The computed weights are then held over the next holding window $[t, t + H)$, where H is the rebalancing step. This rolling train-then-test cycle is repeated through time.

2) No look-ahead bias. Future returns are never used to estimate parameters or choose weights. In other words, the strategy at time t only depends on information available up to $t - 1$.

3) Dynamic portfolio value. If $r_{p,t}$ is the realized portfolio return at date t, the net asset value evolves as:

$$\text{NAV}_t = \text{NAV}_{t-1}(1 + r_{p,t}).$$

This recursion gives the full performance path, not only an average return.

4) Transaction costs and turnover. Rebalancing is not free. Let $\mathbf{w}_t$ be new weights and $\mathbf{w}_{t-1}$ previous weights. Turnover is measured by

$$\tau_t = \sum_{i=1}^{n} |w_{i,t} - w_{i,t-1}|,$$

and a proportional cost is deducted from returns at each rebalance. This prevents overly optimistic results for highly reactive strategies.

5) Relative evaluation. The strategy is compared to a simple benchmark (equal-weight) under the same market period. This allows us to interpret whether complexity actually adds value in terms of return, risk, drawdown, and stability.

Overall, the framework answers a practical question: *If we had followed this rule in real time, with realistic frictions, would performance remain robust out-of-sample?*

Reproducibility and Interactive Dashboard

The report's results are fully reproducible via the Python codebase on GitHub, which can be run with customizable parameters (assets, estimation windows, optimizations). Additionally, an interactive Streamlit dashboard is available in the benchmark_demo folder. To launch it from the terminal, install the dependencies and run the file:

```
pip install -r requirements.txt
streamlit run Cockpit.py
```

The interface allows users to configure the asset universe, constraints, and backtesting period, displaying the efficient frontier, optimal weights, and out-of-sample performance against the two benchmarks in real time.

3.1 Empirical Results on Real S&P 500 Data

Whereas the two illustrative figures above use offline synthetic prices for reproducibility, this section evaluates the *entire* pipeline on **real** market data. We keep every modelling hypothesis of the experiment settings unchanged, daily simple returns, a sector-stratified universe, $\hat{\boldsymbol{\mu}}$ = sample mean, $\hat{\Sigma}$ = EWMA ($\lambda = 0.94$), the max-Sharpe objective solved with `CVXPY`/`CLARABEL` under long-only and a per-asset cap $w_{\max} = 0.15$, an estimation window $W = 252$ days, a holding period $H = 21$ days, transaction costs of 10 bps, and a risk-free rate $r_f = 4\%$. Only the data source changes from synthetic to real, and we add a third, fully passive benchmark: the **S&P 500 index** (`^GSPC`), which represents the opportunity cost of simply buying the market. After requiring a complete 2019–2025 price history, 32 assets are retained; the out-of-sample track record therefore runs from early 2020 (after the first estimation window) and includes the COVID crash and the 2022 rate-tightening drawdown.

Table 1 reports the out-of-sample metrics. Three observations stand out. First, the optimizer does *not* earn a higher return than the naive $1/N$ allocation (CAGR 12.8% vs 12.5%); what it improves is **risk**: lower volatility (21.2% vs 22.8%) and a shallower maximum drawdown (-34.6% vs -40.0%), which lifts the Sharpe ratio from 0.37 to 0.42. This is precisely the channel through which a better Σ adds value. Second, even so, both active and naive equity portfolios remain *below* the capitalization-weighted S&P 500 (Sharpe 0.45), which rode a handful of mega-cap stocks over the period: the index is genuinely hard to beat. Third, turnover is low (3% per rebalance) and no optimization fallback was triggered, confirming the numerical robustness of the convex solver on this universe.

Strategy	**CAGR %**	**Vol %**	**Sharpe**	**Max DD %**	**Turnover**
Markowitz max-Sharpe	12.80	21.16	**0.42**	-34.62	0.03
Equal-weight $1/N$	12.45	22.83	0.37	-40.04	—
S&P 500 index	13.50	20.94	**0.45**	-33.92	—

Table 1: Out-of-sample performance on real S&P 500 data (2020–2025, 32 assets), strategy vs the two benchmarks. Sharpe ratios are net of the 4% risk-free rate and of 10 bps transaction costs.

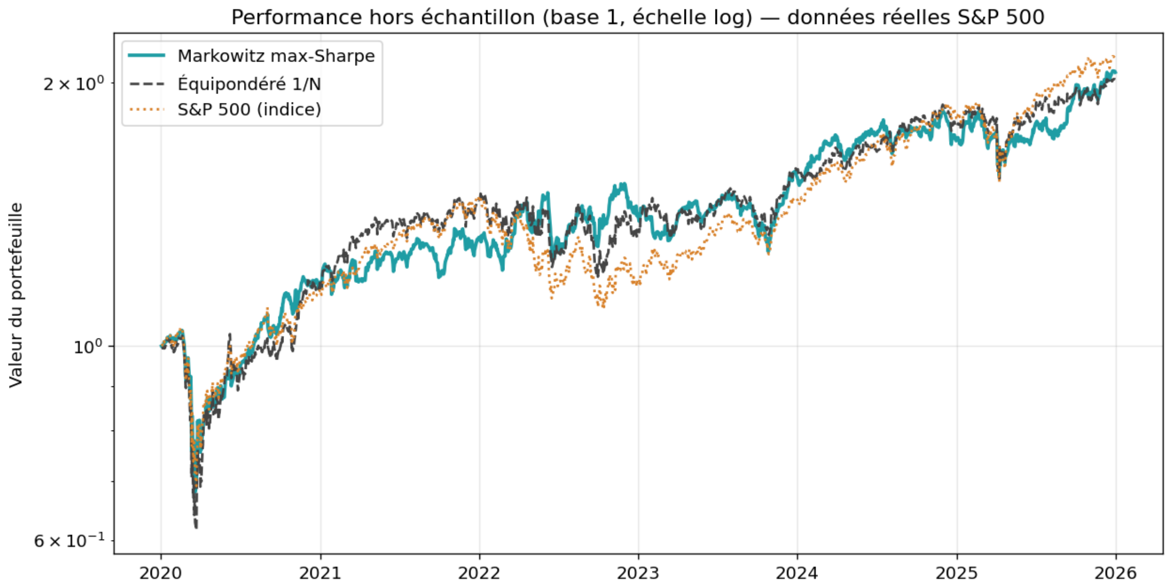

Figure 10: Out-of-sample net asset value (base 1, log scale): Markowitz max-Sharpe vs equal-weight $1/N$ vs the S&P 500 index, on real data. The three paths track closely; the strategy's edge is in risk control, not in raw return.

Figure 11: Drawdown of the strategy through time. The deepest losses coincide with the COVID crash (2020) and the 2022 tightening.

To make the link between drawdowns and *exogenous* market shocks explicit, Figure 12 overlays the two largest real stress episodes of the period (the February–April 2020 COVID crash and the 2022 monetary tightening) on the same value curves. The strategy and both benchmarks lose together during these episodes, illustrating the well-known fact that diversification erodes exactly when correlations spike toward one.

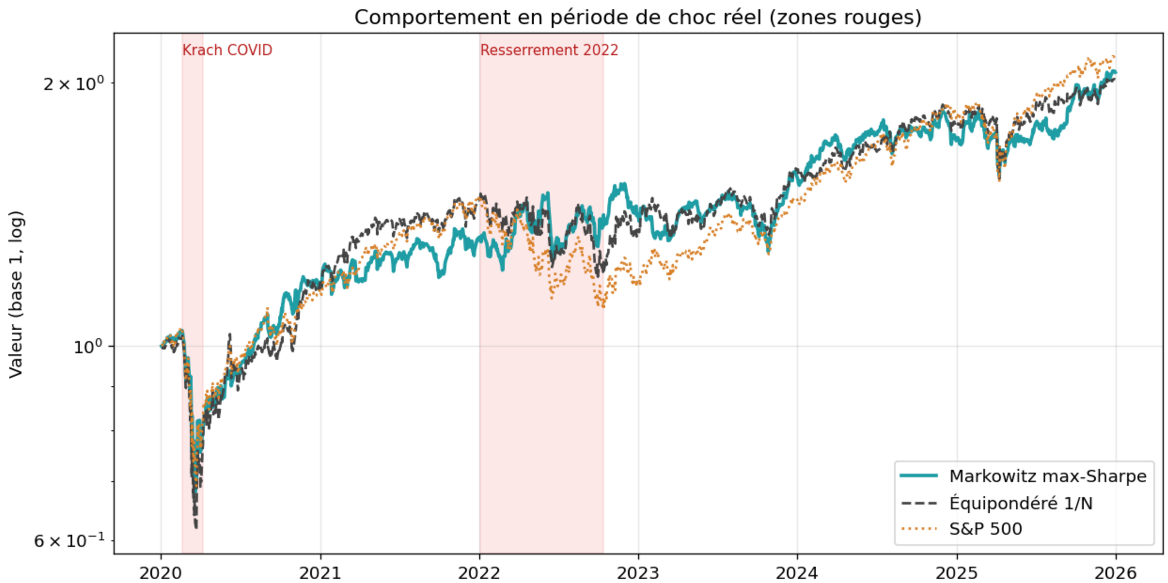

Figure 12: Value paths with the two real shock windows shaded in red. Drawdowns concentrate inside the stress periods, where cross-asset correlations rise and diversification benefits collapse.

Ex-post Validation: do $\hat{\boldsymbol{\mu}}$ and $\hat{\Sigma}$ hold?

A backtest tells us whether a strategy performed well, but not *why*. To diagnose the source of performance, we validate the two statistical ingredients **out-of-sample**: at each rebalancing date we compare the quantity estimated on the past window to the quantity actually realized over the following holding window.

The expected return $\hat{\boldsymbol{\mu}}$ does not hold. The natural metric is the *information coefficient* (IC), the Spearman rank correlation between predicted and realized returns across assets. Figure 13 shows that the IC is statistically indistinguishable from zero for *every* estimator we tried, sample mean, EWMA, James-Stein, rolling mean, and a PCA factor model, with the sample mean at IC ≈ 0.01. The point cloud of predicted versus realized annualized returns (Figure 14) confirms this visually: it is an isotropic blob with a slightly *negative* correlation (-0.06), nowhere near the $y = x$ line of a good forecast. In plain terms, past average returns carry essentially no information about future average returns, the classic finding of DeMiguel et al. (2009), and the reason $\boldsymbol{\mu}$-dependent objectives struggle to beat $1/N$.

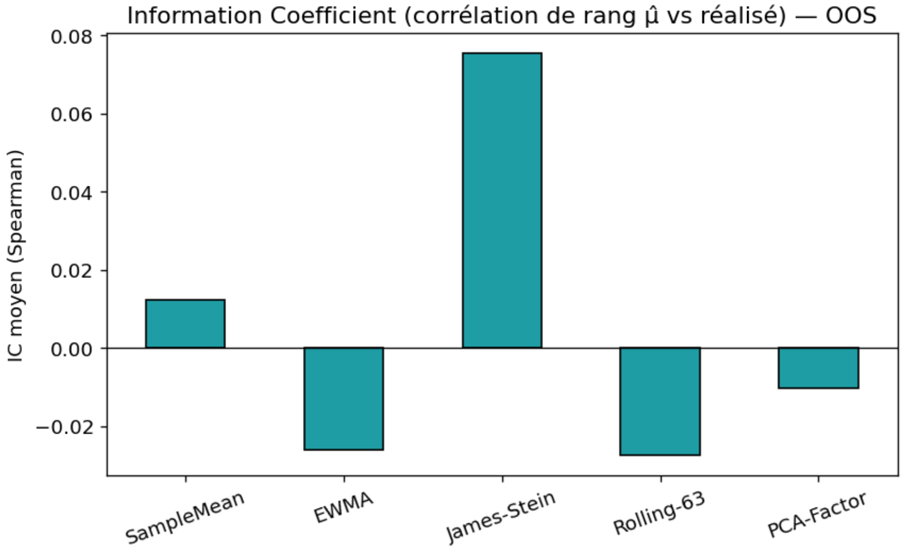

Figure 13: Out-of-sample information coefficient (Spearman rank correlation of $\hat{\mu}$ vs realized returns) for five mean estimators. All are ≈ 0: no estimator predicts the cross-section of returns.

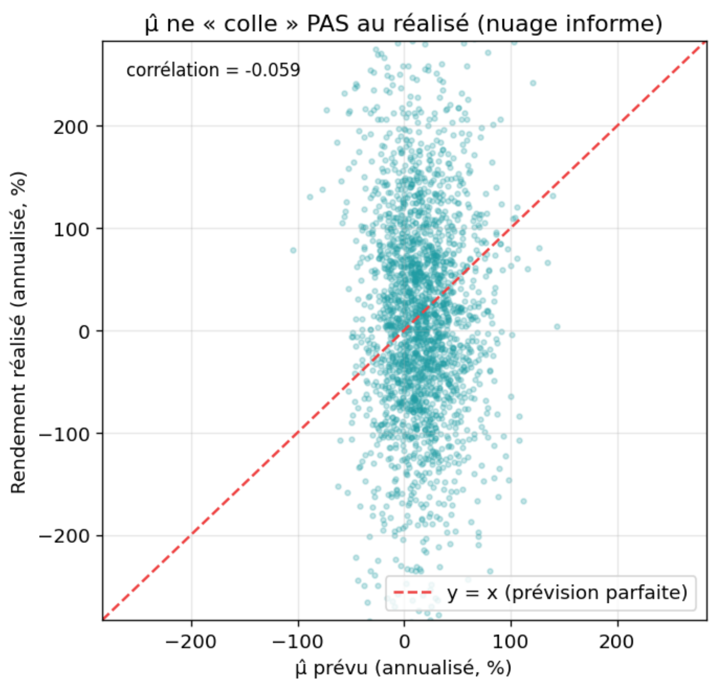

Figure 14: Predicted $\hat{\mu}$ vs realized return (annualized, all assets and windows). The cloud is informationless: $\hat{\mu}$ does *not* stick to the realized return.

The covariance $\hat{\Sigma}$ does hold. The same exercise applied to risk tells the opposite story. Figure 15 ranks the covariance estimators by out-of-sample Gaussian log-likelihood (higher is better) and by absolute volatility-forecast error (lower is better); the regularized estimators (Ledoit-Wolf, EWMA) dominate the raw sample covariance. Crucially, the scatter of predicted versus realized annualized volatility (Figure 16) lines up along the $y = x$ diagonal, with a clearly positive correlation (0.54): the volatility forecast by $\hat{\Sigma}$ is a genuine predictor of realized volatility. This asymmetry, $\hat{\Sigma}$ **is reliable while** $\hat{\boldsymbol{\mu}}$ **is not**, is the central empirical message of the project and explains why the value of optimization lies in risk control rather than return prediction.

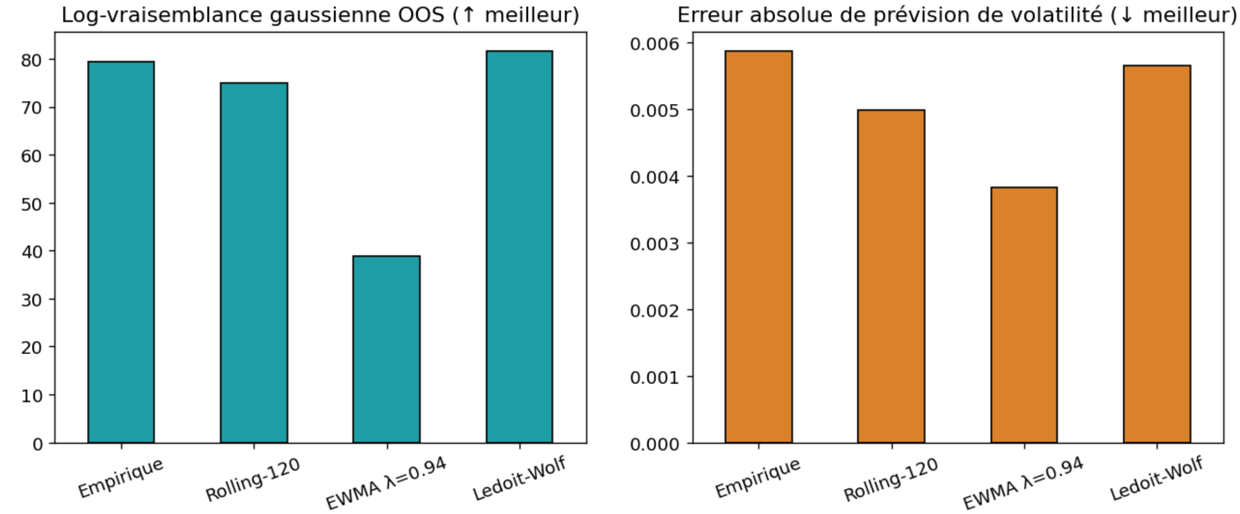

Figure 15: Out-of-sample statistical quality of four covariance estimators: Gaussian log-likelihood (left, ↑) and volatility-forecast error (right, ↓). Shrinkage/EWMA estimators beat the raw sample covariance.

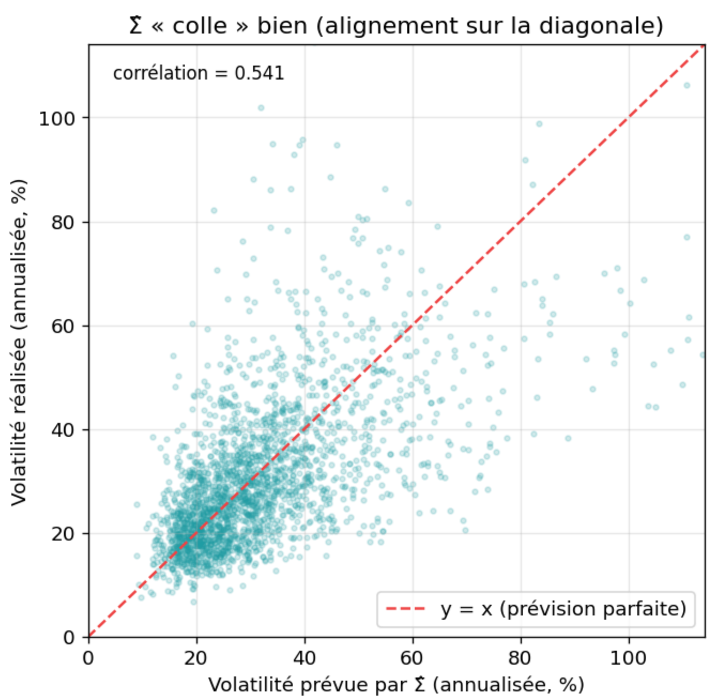

Figure 16: Volatility predicted by $\hat{\Sigma}$ vs realized volatility (annualized, all assets and windows). Points align on the diagonal: $\hat{\Sigma}$ sticks to reality.

Limitations and Assumptions

Our framework, while methodologically careful, rests on a number of assumptions that bound the scope of the conclusions:

- **Gaussian, i.i.d. returns.** The mean-variance objective implicitly assumes that returns are well described by their first two moments and are stable over the estimation window. Part 2 documents that real returns are fat-tailed and asymmetric (excess kurtosis, negative skew), so variance understates tail risk; the VaR/CVaR discussion only partially addresses this.

- **Estimation error dominates, especially on $\boldsymbol{\mu}$.** As shown empirically in the results, the expected-return vector is essentially unpredictable out-of-sample (information coefficient close to zero), which is the main reason $\boldsymbol{\mu}$-dependent objectives (max-Sharpe, mean-variance) fail to beat the naive $1/N$ allocation. Our estimators mitigate but do not eliminate this error.

- **Survivorship bias.** The investable universe is drawn from the *current* composition of the S&P 500. Relative comparisons between methods remain valid, but absolute performance figures are mildly optimistic, since assets that were delisted are absent.

- **Static covariance regime within each window.** Even the EWMA and shrinkage estimators assume a locally stationary covariance over the estimation window; abrupt regime changes (e.g. the COVID crash) are detected only with a lag.

- **Simplified frictions.** Transaction costs are modelled as a proportional charge on turnover; market impact, bid-ask spreads that vary with liquidity, financing costs and taxes are ignored. We also restrict ourselves to long-only, fully-invested, single-currency equity portfolios, without leverage or short positions.

- **Effective universe size.** After requiring a complete price history over the test period, the cleaned universe is smaller than the nominal target; combined with the per-asset weight cap, this limits the room the optimizer has to deviate from equal weighting.

These limitations point directly to the natural extensions discussed in the conclusion: robust (M-)estimation of $\boldsymbol{\mu}$ and Σ, regime-aware covariance prediction, and tail-risk objectives.

Conclusion

This project re-implemented the Markowitz mean-variance framework end to end, from the closed-form and KKT theory of Part 1, through the statistical estimation of its two ingredients in Part 2, to a fully out-of-sample, cost-aware backtest on real S&P 500 data in Part 3. Three conclusions emerge, and they are consistent across the analytical derivations, the simulations and the empirical evidence.

(i) The expected-return vector $\boldsymbol{\mu}$ is, for practical purposes, unpredictable. No estimator, however sophisticated (shrinkage, EWMA, factor models), produces an information coefficient distinguishable from zero out-of-sample, and this holds across estimation windows. **(ii) The covariance matrix Σ, by contrast, is genuinely predictable**: its volatility forecasts line up with realized volatility, and regularized estimators (Ledoit-Wolf, EWMA) improve both conditioning and forecast quality. **(iii) Because mean-variance optimization is most sensitive to errors in $\boldsymbol{\mu}$**, the dimension we estimate worst, the theoretical optimum is fragile, the equally-weighted $1/N$ portfolio is hard to beat, and the value an optimizer can realistically add lies in *risk control* (a better Σ, sensible constraints) rather than in *return forecasting.* On our data the optimized portfolio did beat $1/N$ on a risk-adjusted basis, by lowering volatility and drawdown rather than by raising return, yet still trailed the capitalization-weighted index.

The most promising directions for future work follow directly from these findings and from the limitations above: replacing the sample mean and covariance with *robust M-estimators* that bound the influence of outliers, adopting *regime-aware* covariance predictors of the kind recently proposed by Johansson et al. (2023), and complementing variance with explicit *tail-risk* objectives (CVaR). All of these can be slotted into the modular pipeline and the interactive dashboard developed here without altering the evaluation protocol.

Contribution Section (Not Counted in Word Limit)

The following contributions were made by each team member.

- **Allysa Tan** — created the backtesting engine, implemented performance metrics; wrote the project documentation explaining Markowitz theory and the experimental pipeline
- **Arnaud Michel** Wrote a part of the report on statistic and Markowitz theory, Code development on notebook, figures of part 1 and 2.
- **Matteo Magro** — Statistical metrics in Notebooks, Markowitz Theory and estimators, Bug fixes, contribution to the report
- **Mamadou Kamara** — Markowitz and estimators implementation, notebook initialization, code refactoring and the Streamlit demo, report contibution.
- **Andrea Spinnicchia** — Performance evaluation and global redaction work on the report.
- **Lorenzo Grivettalocia** — Performance evaluation and finalization of the report.

Team workflow and organization tools. We worked through shared notebooks and Python modules in a common Git repository, coordinated tasks through Git commits/branches and iterative review of intermediate results, and synchronized report/code updates continuously.

Use of AI Tools (Not Counted in Word Limit)

In the code. AI assistance was used for boilerplate acceleration, refactoring suggestions, and debugging support. All generated suggestions were reviewed, adapted, and validated by the team before inclusion.

In the report. AI assistance was used for language polishing and structure suggestions. Technical content, methodological choices, equations, and final conclusions were verified and approved by the authors.

Bibliography

References

[1] Attilio Meucci. *Risk and Asset Allocation*. Springer

[2] Capinski, M., & Zastawniak, T. (2010). *Mathematics for Finance: An Introduction to Financial Engineering*. Springer, pp. 53–89.

[3] Sabourin, A., Fercoq, O., & Portier, F. *Analyse numérique, notes de cours (MACS 205)*. Télécom Paris.

[4] Charon, I., & Hudry, O. *Analyse numérique et optimisation continue, notes de cours (FMA0EL01TP)*. Télécom Paris.

[5] Markowitz, H. (1952). *Portfolio Selection*. Journal of Finance, 7(1), 77–91.

[6] Boyd, S., Johansson, K., Kahn, R., Schiele, P., & Schmelzer, T. (2024). *Markowitz Portfolio Construction at Seventy*. arXiv:2401.05080.

[7] Johansson, K., Ogut, M. G., Pelger, M., Schmelzer, T., & Boyd, S. (2023). *A Simple Method for Predicting Covariance Matrices of Financial Returns*. Foundations and Trends in Econometrics, 12(4), 324–407. DOI: 10.1561/0800000047.

[8] Palomar, D. P. (2025). *Portfolio Optimization: Theory and Application*. Cambridge University Press.

[9] Ledoit, O., & Wolf, M. (2004). *A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices*. Journal of Multivariate Analysis, 88(2), 365–411.

[10] Ledoit, O., & Wolf, M. (2004). *Honey, I Shrunk the Sample Covariance Matrix*. Journal of Portfolio Management, 30(4), 110–119.

[11] James, W., & Stein, C. (1961). *Estimation with Quadratic Loss*. Proc. 4th Berkeley Symposium on Mathematical Statistics and Probability, 1, 361–379.

[12] DeMiguel, V., Garlappi, L., & Uppal, R. (2009). *Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy?* Review of Financial Studies, 22(5), 1915–1953.

[13] Chopra, V. K., & Ziemba, W. T. (1993). *The Effect of Errors in Means, Variances, and Covariances on Optimal Portfolio Choice*. Journal of Portfolio Management, 19(2), 6–11.

[14] J.P. Morgan/Reuters (1996). *RiskMetrics — Technical Document*, 4th ed. New York.

[15] Rockafellar, R. T., & Uryasev, S. (2000). *Optimization of Conditional Value-at-Risk*. Journal of Risk, 2(3), 21–41.

[16] Boyd, S., & Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press.